

EXHIBIT 18

Ex. 18 – Infringement Contention Claim Chart
U.S. Patent No. 12,288,421 (“the ’421 patent”)

Apple’s products directly and/or indirectly infringe at least claims 1, 5, 6, 7, 8, 9, 11, 17, 20, 21, 23, 24 (“Asserted Claims”) of U.S. Patent No. 12,288,421 (“the ’421 patent”). The products that infringe the ’421 patent include at least the iPhone 15, iPhone 15 Plus, iPhone 15 Pro, iPhone 15 Pro Max, iPhone 16, iPhone 16e, iPhone 16 Pro, iPhone 16 Pro Max, iPhone 16 Plus, iPhone 17, iPhone 17e, iPhone 17 Pro, iPhone 17 Pro Max, iPhone Air, 11-inch iPad Pro (4th Generation), iPad Pro 12.9 inch (6th Generation), iPad Pro 11-inch (M4 and M5), iPad Pro 13-inch (M4 and M5) and other infringing iPhone and iPad models (each an “Accused Apple Product” and collectively the “Accused Apple Products”). The Accused Apple Products infringe the ’421 patent as set forth below. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

Additionally, the Apple iPhone 17 is representative of the Accused Apple Products at least because the other Accused Apple Products function in the same or substantially similar manner to the Apple iPhone 17.¹ The relevant identification processes and functionalities, as well as the hardware components of the Apple iPhone 17, are demonstrated through testing of the Apple iPhone 17 (“Apple Product Testing”), which is available for inspection upon request.

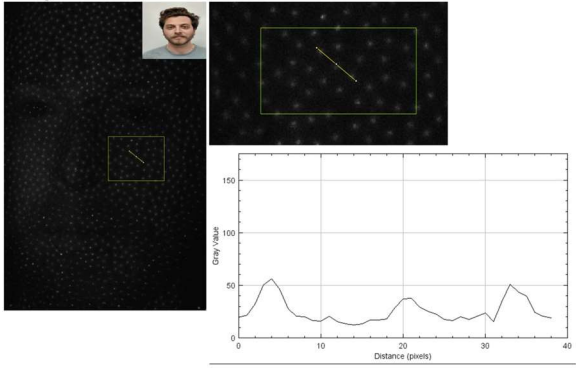
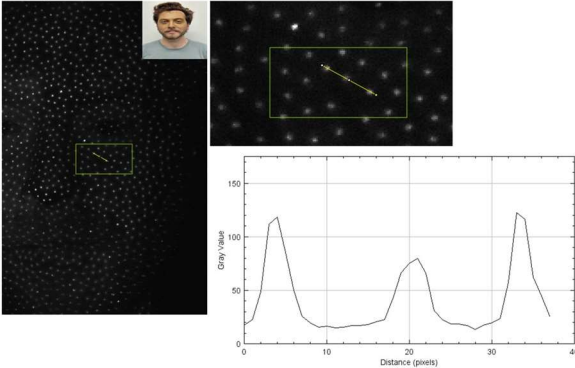
Claim 1	Accused Apple Products
[1pre] A method for face authentication comprising the following steps:	The Accused Apple Products comprise a method for face authentication. “With a simple glance, Face ID securely unlocks supported Apple devices.” Exhibit 35, at 22.
[1a1] at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining	The Accused Apple Products comprise at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps” Exhibit 35, at 22.

¹ Each of the Accused Apple Products includes the claimed hardware components and steps. The hardware components and source code implementing the relevant functionalities of the Apple iPhone 17 are representative of the Accused Apple Products for the purpose of infringement. To the extent there may be differences in the hardware composition, certain software steps, or source code versions among these Accused Apple Products, those differences are not material for the infringement analysis.

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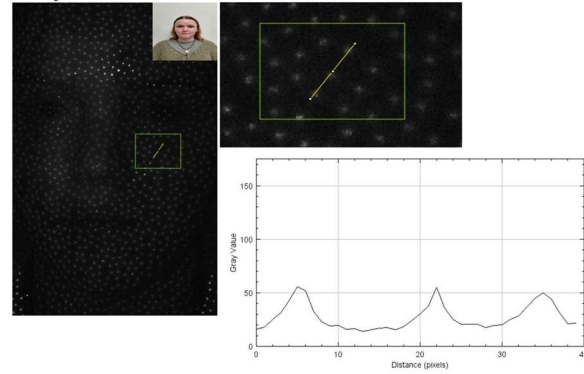
at least one second image using the at least one camera,	The iPhone 17 comprises a “dot projector + flood illuminator + NIR camera module.” Exhibit 34C, at 26. The dot projector VCSEL projects the infrared dot pattern, and the near-infrared camera captures the reflected pattern using a monochrome NIR CMOS image sensor. Exhibit 34C 27, 41, 47.
[1a2] wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features,	The Accused Apple Products comprise the second image comprising a plurality of reflection features generated by the scene in response to illumination by the illumination features. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps” Exhibit 35, at 22. For example, Face ID on the Apple iPhone 17 captures features reflected by dots projected onto the scene. Subject01 Covered area: none – natural Face Subject01 Covered area: entire face

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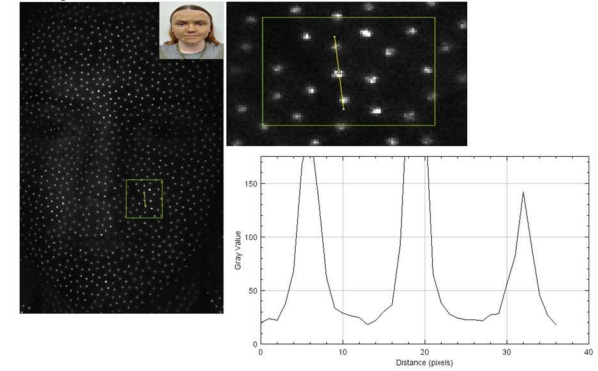
	Subject02 Covered area: none – natural Face  Subject02 Covered area: entire face  Exhibit 47 ¶¶ 23-31.
[1a3] wherein the skin detection step comprises determining at least one material property of the reflection features by using the processing unit,	The Accused Apple Products comprise the skin detection step comprising determining at least one material property of the reflection features by using the processing unit. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29. For example, Face ID on the Apple iPhone 17 responds differently to different material properties of the reflection features, using the processing unit to do so.

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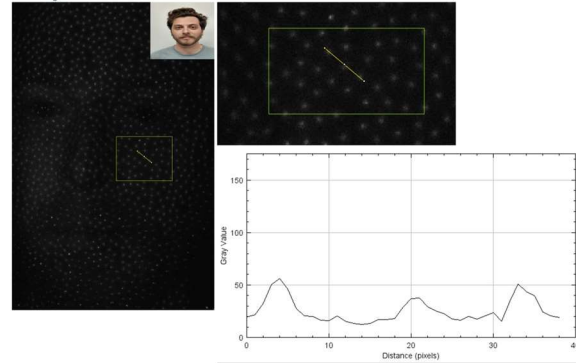
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

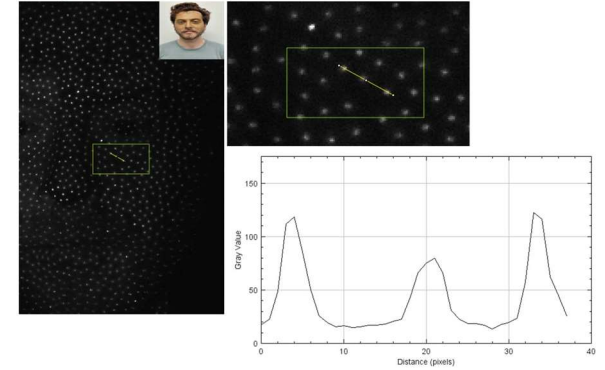


Exhibit 47 ¶¶ 23-31.

Additionally and alternatively, Face ID works “even with a mask on,” indicating that Face ID can distinguish skin from non-skin material.

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Face ID security

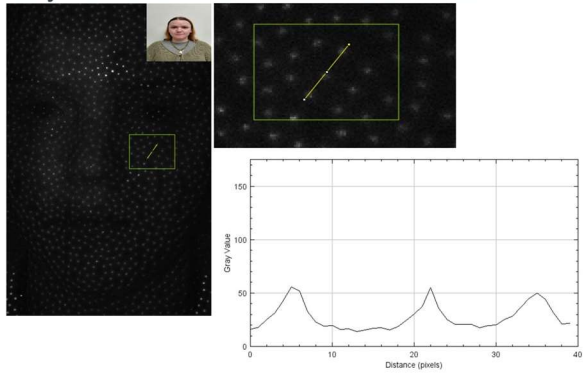
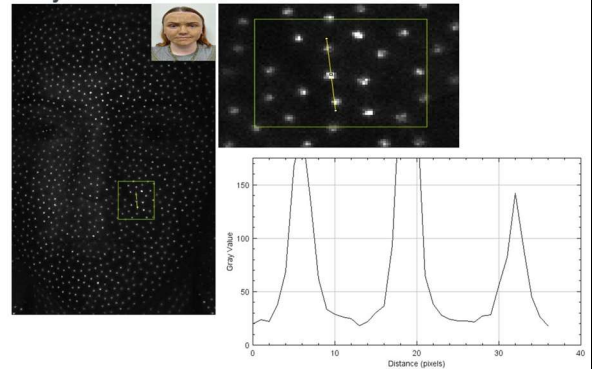
With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

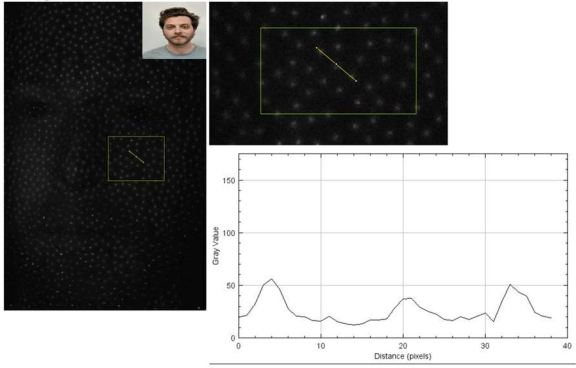
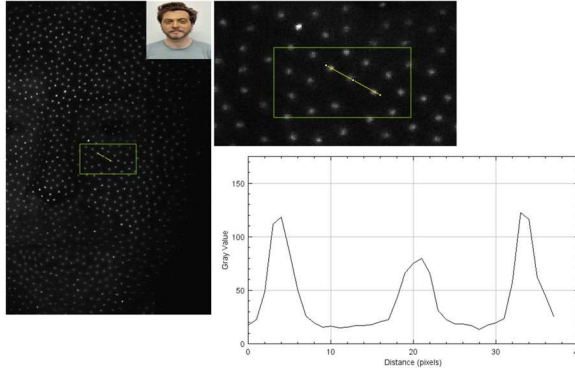
The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

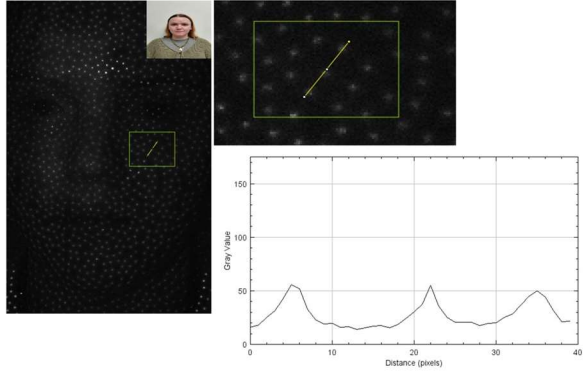
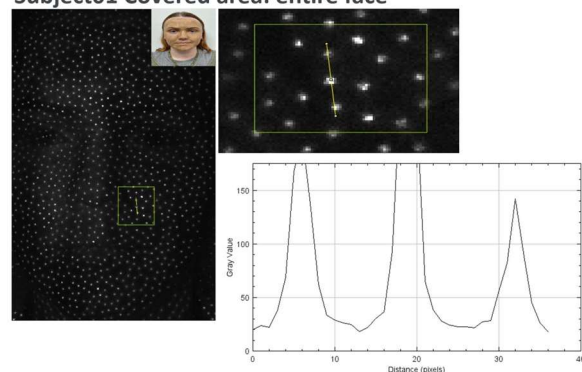
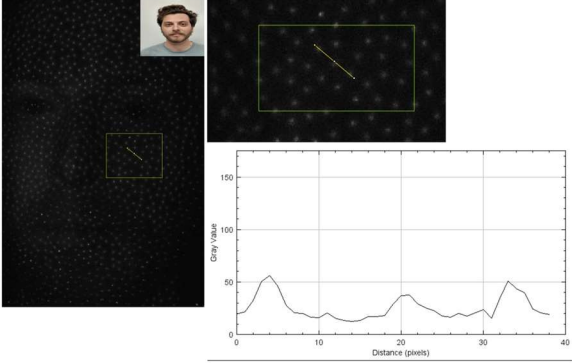
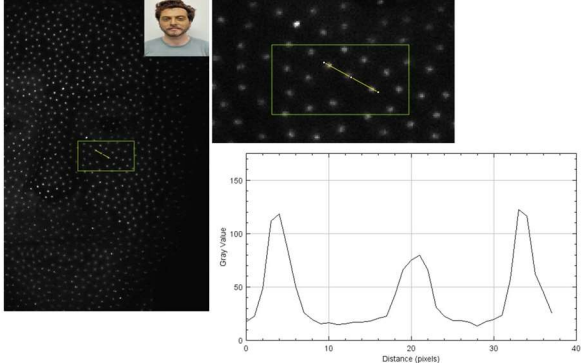
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	Exhibit 35, at 22.
[1a4] wherein a detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin,	The Accused Apple Products comprise a detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29. For example, Face ID on the Apple iPhone 17 responds in one way (by unlocking the phone) when a detected face is characterized as skin based on material properties characteristic of skin and responds in another way (by denying the unlock attempt) when a detected face is characterized by non-skin material properties. Subject01 Covered area: none – natural Face  This figure shows a visual analysis of a natural face. It includes a small inset photo of a woman, a main image of a face with a green bounding box, and a line graph of Gray Value vs. Distance (pixels). The graph shows three distinct peaks, with the highest peak reaching approximately 100 gray value units. Subject01 Covered area: entire face  This figure shows a visual analysis of a face where the entire area is covered. It includes a small inset photo of a woman, a main image of a face with a green bounding box, and a line graph of Gray Value vs. Distance (pixels). The graph shows three distinct peaks, with the highest peak reaching approximately 150 gray value units.

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	Subject02 Covered area: none – natural Face  Subject02 Covered area: entire face  Exhibit 47 ¶¶ 23-31.
[1a5] wherein the illumination pattern is selected from the group consisting of at least one point pattern, at least one line pattern, at least one stripe pattern, at least one checkerboard pattern, and at least one pattern comprising an arrangement of periodic or non-periodic features;	The Accused Apple Products comprise the illumination pattern is selected from the group consisting of at least one point pattern, at least one line pattern, at least one stripe pattern, at least one checkerboard pattern, and at least one pattern comprising an arrangement of periodic or non-periodic features. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps” Exhibit 35, at 22. For example, the iPhone 17 comprises a “dot projector + flood illuminator + NIR camera module.” Exhibit 34C, at 26. The dot projector VCSEL projects the infrared dot pattern, and the near-infrared camera captures the reflected pattern using a monochrome NIR CMOS image sensor. Exhibit 34C 27, 41, 47. For example, Face ID on the Apple iPhone 17 uses at least the following illumination pattern.

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	Subject01 Covered area: none – natural Face  Subject01 Covered area: entire face  Subject02 Covered area: none – natural Face  Subject02 Covered area: entire face  Exhibit 47 ¶¶ 23-31.
[1b] at least one 3D detection step, wherein the 3D detection step comprises determining at least one depth level from the reflection features of the second image,	The Accused Apple Products comprise at least one 3D detection step, wherein the 3D detection step comprises determining at least one depth level from the reflection features of the second image, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level, wherein the 3D detection step comprises determining depth information of the reflection features by structured light.

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wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level, wherein the 3D detection step comprises determining depth information of the reflection features by structured light; and	“With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.” Exhibit 35, at 22. “Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
[1c] at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step a) the detected face is characterized as	The Accused Apple Products comprise at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step a) the detected face is characterized as skin and in step b) the detected face is characterized as 3D object. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides

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skin and in step b) the detected face is characterized as 3D object.	intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.” Exhibit 35, at 22. “Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
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